

DBMS Semester Examination 2023

Database Management Systems (DBMS) · Semester Examination · 2023

Max. Marks	75	Time	3 hours
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Answer ALL questions from Part A and any FIVE from Part B.

Part A (25 marks)

1. Discuss the role of Database System Architecture in Introduction & ER Model. (5 marks)
2. Write short notes on Relational Algebra. (5 marks)
3. Illustrate Functional Dependencies with a worked example. (5 marks)
4. Discuss the role of ACID Properties in Transactions & Concurrency Control. (5 marks)
5. Explain File Organization with a neat diagram and a suitable example. (5 marks)

Part B

1. What is meant by Database System Architecture? Discuss its advantages and limitations. (Unit 1)
2. Illustrate Data Independence with a worked example. (Unit 1)
3. Compare Relational Algebra with Joins. When would you prefer one over the other? (Unit 2)
4. Define Joins. Derive or justify the key properties involved. (Unit 2)
5. Discuss the role of Functional Dependencies in Normalization & Schema Refinement. (Unit 3)
6. Define Closure of Attributes. Derive or justify the key properties involved. (Unit 3)
7. Explain ACID Properties with a neat diagram and a suitable example. (Unit 4)
8. Explain Serializability with a neat diagram and a suitable example. (Unit 4)
9. Explain File Organization with a neat diagram and a suitable example. (Unit 5)
10. Write short notes on B+ Tree Indexes. (Unit 5)